

Chapter 8 Individual Assignment Report

Employee Directory with Vue 3, Axios, Express and MySQL

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1. Working User Interface

The application provides a single-page HR directory with an add/edit form, active headcount, server-backed search and sort controls, and a responsive employee table. Active and inactive employees are visually separated through status badges, and salaries are formatted in Malaysian Ringgit.

CHAPTER 8 INDIVIDUAL ASSIGNMENT

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Employee Directory

A single-page HR tool built with Vue 3, Axios, Express and MySQL.

NEW EMPLOYEE

Add Employee

Employee ID

EMP108

Full Name

Employee full name

Email

name@myhrpilot.test

Department

Select department

Position

Job title

Hire Date

mm/dd/yyyy

Salary (RM)

4500

☒ Currently active employee

Add Employee

DIRECTORY STATUS

6 active of 7 total

Search

Name, ID, email, or department

Sort by

Employee ID

Order

Ascending

Clear

EMPLOYEE RECORDS

Directory List

EMPLOYEE	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
Lina Tan Mei Ling EMP101 - lina.tan@myhrpilot.test	HR	People Operations Executive	12 Apr 2021	RM 4,300.00	Active	<button>Edit</button> <button>Delete</button>
Daniel Lim Wei Jian EMP102 - daniel.lim@myhrpilot.test	IT	Systems Analyst	28 Sept 2020	RM 6,900.00	Active	<button>Edit</button> <button>Delete</button>
Farah Nadia Ismail EMP103 - farah.ismail@myhrpilot.test	Finance	Accounts Officer	07 Feb 2022	RM 5,100.00	Active	<button>Edit</button> <button>Delete</button>
Jason Goh Kai Sheng EMP104 - jason.goh@myhrpilot.test	Marketing	Brand Coordinator	19 Jun 2023	RM 3,900.00	Active	<button>Edit</button> <button>Delete</button>
Mira Aisyah Rahman EMP105 - mira.rahman@myhrpilot.test	Operations	Logistics Supervisor	04 Nov 2019	RM 6,200.00	Active	<button>Edit</button> <button>Delete</button>

Figure 1: Desktop layout showing the employee form, headcount, search/sort controls, and loaded records.

Additional Responsive Screenshot

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Figure 2: Tablet-width layout with the same controls stacked safely for narrower screens.

2. Learning Outcome Coverage

Learning Outcome	How the project satisfies it
LO1: Connect Vue to backend	App.vue loads employees from the Express API and refreshes state after create, update, delete, search, and sort actions.
LO2: Use Axios	src/api/employeeApi.js defines one Axios instance with baseURL, timeout, JSON headers, request logging, response error
LO3: Vue forms and validation	EmployeeForm.vue uses v-model.trim and v-model.number, validates all required fields, shows inline errors, and emits sav
LO4: CRUD with MySQL	server/index.js implements GET, POST, PUT, and DELETE routes using mysql2/promise prepared statements, plus server

3. Technical Decisions

The project uses a single root package.json so the grader only needs to run npm install once. The backend stays inside the server folder, while Vue components are split into EmployeeForm, SearchSort, and EmployeeList. The MySQL pool uses dateStrings: true so hireDate remains stable as YYYY-MM-DD in the frontend.

Prepared statements are used for all dynamic SQL values. Sorting is handled by mapping client sort keys to a server-side whitelist, preventing arbitrary column names from being injected into ORDER BY.

4. Challenges and Resolutions

Challenge	Resolution
Preventing unsafe SQL sorting	Only known sort keys are accepted and mapped to fixed column names before building ORDER BY.
Keeping dates consistent	The MySQL connection is configured with dateStrings: true, avoiding timezone shifts in hireDate values.
Handling validation clearly	Client-side validation blocks invalid submits, while backend validation returns structured field errors for duplica
Making the interface usable on tablets	Controls stack at narrower widths and the employee table uses horizontal scrolling to avoid overlapping colum

The application was verified with npm install and npm run build. Database import instructions are included in README.md for Laragon/MySQL setup before running the final app.